

VIJAY RAKESH REDDY BANDELA

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SUMMARY

Detail-oriented Data Engineer with 3+ years of expertise in designing scalable data pipelines, optimizing cloud storage, and delivering real-time analytics for healthcare and financial sectors. Skilled in Apache Kafka, Apache Spark, PySpark, AWS, Snowflake, and Tableau, driving efficient data processing and integration. Improved data processing efficiency by up to 40% and reduced data retrieval times by 25%. Led successful data migration projects, ensuring HIPAA and GDPR compliance, and automated workflows for increased operational efficiency. Developed predictive models and real-time dashboards, contributing to cost savings and better decision-making. Experienced in managing post-merger data integrations, working closely with cross-functional teams, and supporting strategic initiatives, such as cost-benefit analysis for biosimilar alternatives. Strong in project management, Agile methodologies, and delivering actionable insights to stakeholders. Proven ability to optimize healthcare data systems, reduce manual handling, and enhance service delivery through data-driven solutions.

EDUCATION

Master of Science in Computer Science | University of Alabama at Birmingham December 2024
Bachelors of Technology in Electronics and Communication Engineering | SRM Institute of Science and Technology May 2022

PROFESSIONAL EXPERIENCE

United Health Group, Boston, MA, USA | Data Engineer Jan 2024-Present

- Utilized AWS Glue, Snowflake, and PySpark to process socioeconomic and healthcare datasets, enabling advanced predictive analytics to assess the \$5.1M Boston housing investment. Projected a 15% improvement in community well-being metrics, showcasing the alignment of this initiative with UHG’s social impact goals.
- Engineered data pipelines using Kafka and AWS Redshift to integrate operational and financial datasets from UHG and Amedisys, ensuring seamless post-merger readiness. Focused on maintaining compliance with regulatory frameworks and minimizing operational disruptions during the extended merger timeline.
- Designed cost-benefit analysis models in PySpark and Snowflake to assess the transition from Humira to biosimilar alternatives, such as Amjevita. These models projected \$50M in annual cost savings while ensuring that member affordability and healthcare outcomes remained a top priority.
- Implemented real-time dashboards in Tableau integrated with AWS Redshift, providing stakeholders with up-to-date insights into the financial and operational impacts of these strategic developments. Enhanced data accessibility improved decision-making processes across teams.
- Automated data transformation workflows using AWS Glue to streamline reporting related to housing investment impacts, enabling the delivery of actionable insights to policymakers and community health teams.
- Collaborated with cross-functional teams to develop member communication strategies based on data-driven insights, ensuring a smooth transition to biosimilar reimbursement plans and reinforcing UHG’s focus on cost-effective healthcare solutions.

Magna Infotech, Bangalore, India | Data Engineer Jan 2021 – Dec 2022

- Designed and implemented scalable data pipelines using Apache Kafka and Apache Spark, improving real-time data processing efficiency by 30%, ensuring faster data flow and reduced processing latency.
- Optimized cloud data storage solutions on AWS, achieving HIPAA and GDPR compliance for healthcare data, enhancing security and scalability, and reducing data retrieval times by 25%.
- Engineered data migration of critical pharmacy benefit management (PBM) data to Hadoop clusters and Cassandra databases, improving data processing capabilities by 40%, enabling faster analytics and reporting.
- Developed ETL processes for integrating data across healthcare platforms, reducing manual data handling by 25%, enabling real-time reporting and improving operational efficiency.
- Collaborated with cross-functional teams to implement data solutions for Medicare Advantage programs, resulting in improved service delivery and a 10% increase in operational efficiency.
- Improved predictive analytics models leading to a 12% reduction in healthcare costs and better forecasting accuracy.

PROJECTS

Amedisys Merger Data Integration (United Health Group)

- Designed and implemented scalable data pipelines using Kafka and AWS Redshift to consolidate operational and financial datasets from UHG and Amedisys, ensuring smooth post-merger integration.
- Extracted and integrated data from various sources across both companies, including financial records, healthcare service data, and operational metrics, aligning the datasets for seamless integration.
- Developed real-time data pipelines using Kafka for data streaming and AWS Redshift for efficient data storage and querying, ensuring continuous data flow between both companies’ systems.
- Applied data transformation and cleansing techniques to standardize and prepare the datasets, ensuring data integrity for compliance and reporting purposes.
- Conducted rigorous testing and validation to ensure that the data pipelines met regulatory requirements and internal governance standards, preventing data-related compliance issues.
- Delivered consolidated datasets ready for post-merger financial analysis and reporting, reducing operational delays by 20% and enabling timely decision-making, thus ensuring the merger progressed without significant setbacks.

Real-Time Healthcare Data Integration and Analytics Platform (Magna Infotech)

- Integrated diverse healthcare data sources into a unified, real-time platform, enabling predictive analytics and efficient reporting.
- Designed and implemented Apache Kafka and Apache Spark-based data pipelines, improving real-time data processing efficiency by 30%, ensuring faster data flow and reduced latency.
- Engineered AWS-based, HIPAA and GDPR-compliant cloud storage solutions, enhancing data security and scalability, while reducing data retrieval times by 25%.
- Led the migration of critical pharmacy benefit management (PBM) data to Hadoop clusters and Cassandra databases, improving data processing capabilities by 40%, and enabling real-time analytics.
- Developed and automated ETL processes to integrate data across Medicare, Medicaid, and virtual care platforms, reducing manual data handling by 25% and improving reporting accuracy and timeliness.
- Collaborated with cross-functional teams to implement data solutions for Medicare Advantage programs, resulting in a 10% improvement in operational efficiency and service delivery.
- Optimized predictive analytics models for healthcare data, leading to a 12% reduction in patient readmissions and improving healthcare cost forecasting.

TECHNICAL SKILLS

Operating Systems: Windows, Linux, Unix
Programming & Data Processing: Python (Pandas, NumPy), SQL, ETL Processes, Data Cleaning, Data Preprocessing, Missing Data Imputation, Data Transformation, Data Aggregation
Real-Time Data Processing: Apache Kafka, Apache Spark Streaming, Streaming Delta Live Tables (DLT), PySpark
Cloud Technologies & Data Storage: AWS (Glue, Lambda, RedShift), Azure (Delta Lake), Hadoop, Cassandra
Data Integration & Migration: Data Lakes, Data Partitioning, Data Ingestion, Data Validation Automation, Data Migration
Data Visualization: Tableau, Power BI, Dashboards, Real-Time Data Visualizations, Trend Analysis, KPI Monitoring
Statistical Analysis & Modeling: Predictive Analytics, Statistical Methods, Trend Identification, Correlation Analysis
Database Management & Querying: SQL Databases, Data Extraction, Query Optimization, Data Validation Workflows
Automation & Scripting: Python (Automation, Anomaly Detection), Reporting Automation, Real-Time Data Ingestion, Data Pipeline Development
Data Security & Compliance: HIPAA Compliance, GDPR Compliance, Data Privacy Best Practices, Encryption, Secure Data Handling